ANALYSING UNIVERSITY GRADUATION PATTERNS AND ACADEMIC PERFORMANCE TRENDS. (A CASE STUDY OF THE UNIVERSITY OF NIGERIA, NSUKKA 2024 GRADUATION RECORDS)

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DATA ANALYTICS

TOOLS:

MICROSOFT EXCEL | POWER QUERY | DAX | POWER BI

CHAPTER ONE

1.1 Introduction

Higher education institutions rely on graduate data to evaluate academic performance, monitor student outcomes, and improve educational planning. By analyzing graduation records, universities can identify trends in student performance, demographic characteristics, and completion rates that support better academic decision-making.

This project analyzes the 2024 University of Nigeria (UNN) graduation dataset using Excel and Power BI to uncover patterns in academic performance, gender distribution, geographic representation, and graduation outcomes. The findings provide insights that can help improve student support services and institutional planning.

1.2 Problem Statement

Universities generate large amounts of student data every academic session, but much of this information is underutilized. Without proper analysis, it becomes difficult to identify high-performing faculties, monitor graduation trends, understand demographic patterns, or make informed academic decisions.

This project addresses that challenge by analyzing graduation records to provide meaningful insights for university management.

1.3 Aim of the Study

The aim of this project is to analyze university graduation records and identify patterns in academic performance, demographic characteristics, and graduation outcomes using data analytics techniques.

1.4 Objective

The objectives of this study are to:

- Analyze the distribution of graduates across faculties.
- Examine academic performance based on honors obtained.
- Analyze gender distribution among graduates.
- Identify the states that contribute the highest number of graduates.
- Compare academic performance across faculties.
- Provide recommendations for improving academic planning

1.5 Scope

The study is limited to the University of Nigeria 2024 graduation dataset and focuses on graduation records, demographic information, academic performance, and faculty distribution.

1.6 Tools Used

- Microsoft Excel
- Power BI

CHAPTER TWO

LITERATURE REVIEW

Educational institutions increasingly use data analytics to improve teaching, learning, and administrative decision-making. Academic data analysis enables universities to evaluate student performance, monitor graduation rates, and identify factors influencing academic success.

Business Intelligence tools such as Power BI allow institutions to transform raw educational data into interactive dashboards that support evidence-based decision-making. Similarly, Data analytics tool like Excel provides efficient tools for data cleaning, statistical analysis and visualization.

Combining these technologies enables universities to identify trends, improve student support services, and allocate resources more effectively.

CHAPTER THREE

METHODOLOGY

The analysis followed five major stages:

Data Collection

The graduation dataset containing student demographic and academic information was provided by the department.

Data Cleaning

The following cleaning activities were performed:

- Handled missing values
- Standardized Faculty & Department names
- Checked duplicates
- Verified data inconsistency
- Standardized State names
- Created Completion Status
- Created Region
- Created Honor Code
- Created Age Group
- Converted date column into the correct format

Data Analysis

The cleaned data was analyzed using Power BI and Python to answer key research questions.

Visualization

Interactive dashboards were created using:

- ❖ KPI Cards
- ❖ Bar Charts
- ❖ Pie Charts
- ❖ Doughnut Charts
- ❖ Column Charts
- ❖ Line Charts
- ❖ Slicers

CHAPTER FOUR

4.1 Key Findings

1. Academic Performance Across Faculties

The analysis revealed noticeable differences in academic performance across the various faculties. Most graduates earned **Second Class Honours (Lower Division)**, making it the most common degree classification. In contrast, **First Class Honours** accounted for only a small percentage of the graduating population. A few faculties consistently recorded a higher proportion of First Class graduates, indicating stronger academic performance within those programmes.

Insight: Academic achievement varies across faculties, with a small number of faculties producing a larger share of top-performing graduates.

2. Gender Distribution

The analysis showed that female graduates made up a larger proportion of the graduating class than male graduates. Female students accounted for approximately **60.7%** of graduates, while male students represented about **39.3%**.

Insight: The results suggest strong female participation in higher education within the university.

3. Faculty Distribution

The **Faculty of Social Sciences** recorded the highest number of graduates, followed by faculties such as **Faculty of Art** and **Faculty of Biological Science**. Other faculties had considerably smaller graduating classes.

Insight: The variation in graduate numbers across faculties may reflect differences in student enrolment, programme demand, and available resources.

4. Geographic Distribution

Most graduates came from states within the **South-East geopolitical zone**, particularly **Enugu** and **Anambra** States. Comparatively fewer graduates originated from states in other regions of the country.

Insight: The university attracts a significant proportion of students from neighbouring states, presenting opportunities to strengthen recruitment efforts in other geopolitical zones.

5. Age Distribution

The majority of graduates were between **22 and 27 years old** at the time of graduation. Only a small number of students graduated below the age of 22 or above 28.

Insight: The age profile indicates that most students completed their undergraduate programmes within the expected age range.

6. Programme Completion Rate

Approximately **77%** of students completed their programmes within the expected duration, while the remaining students graduated after spending additional years in school.

Insight: Although the overall completion rate is encouraging, targeted support for students experiencing delays could further improve graduation outcomes.

7. Academic Performance by Gender

Both male and female students obtained all categories of academic honours. However, the overall distribution of degree classifications was similar for both genders, indicating no significant difference in academic performance based on gender.

Insight: Academic success appears to be influenced more by programme or faculty than by gender.

8. Relationship Between Programme Duration and Years Spent

The analysis indicated a moderate positive relationship between **Course Duration** and **Years Spent**, meaning that students enrolled in longer programmes generally remained in school for more years. Age showed only a weak relationship with these variables.

Insight: Programme duration influences the total time spent in school, while graduation age is also affected by other factors such as admission age and delayed completion.

4.2 Summary of Findings

The analysis produced the following key results:

- Total Graduates: **5,347**
- Total Faculties: **19**
- Total Departments: **85**

- Female graduates outnumbered male graduates.
- **Second Class Honours (Lower Division)** was the most frequently awarded degree classification.
- First Class graduates represented a relatively small proportion of the graduating class.
- Approximately **77%** of students completed their programmes within the expected duration.
- Most graduates were between **22 and 27 years** of age.
- **Enugu** and **Anambra** States recorded the highest number of graduates.
- The **Faculty of Social Sciences** had the largest graduating class.
- A moderate positive relationship was observed between programme duration and years spent in school.

CHAPTER FIVE

RECOMMENDATIONS

Recommendation 1

The university should encourage collaboration between high-performing and lower-performing faculties by promoting the sharing of effective teaching methods, mentoring programmes, and academic support strategies.

Recommendation 2

Faculties with lower on-time completion rates should strengthen academic advising, early intervention programmes, and student support services to reduce delayed graduation.

Recommendation 3

To improve geographic diversity, the university should increase awareness and recruitment activities in states and regions with lower student representation.

Recommendation 4

University management should make greater use of interactive dashboards and data analytics to continuously monitor academic performance, completion rates, and student demographics, enabling timely and evidence-based decisions.

CONCLUSION

This study successfully examined the University of Nigeria graduation records using **Microsoft Excel** for data preparation and **Microsoft Power BI** for analysis and visualization. The findings highlighted important trends in academic performance, demographic characteristics, geographic representation, and programme completion.

The interactive dashboard developed during this project demonstrates how educational data can be transformed into meaningful insights that support academic planning, resource allocation, and student support initiatives. Overall, the study shows that business intelligence tools such as Power BI play an important role in helping higher education institutions make informed, data-driven decisions.